

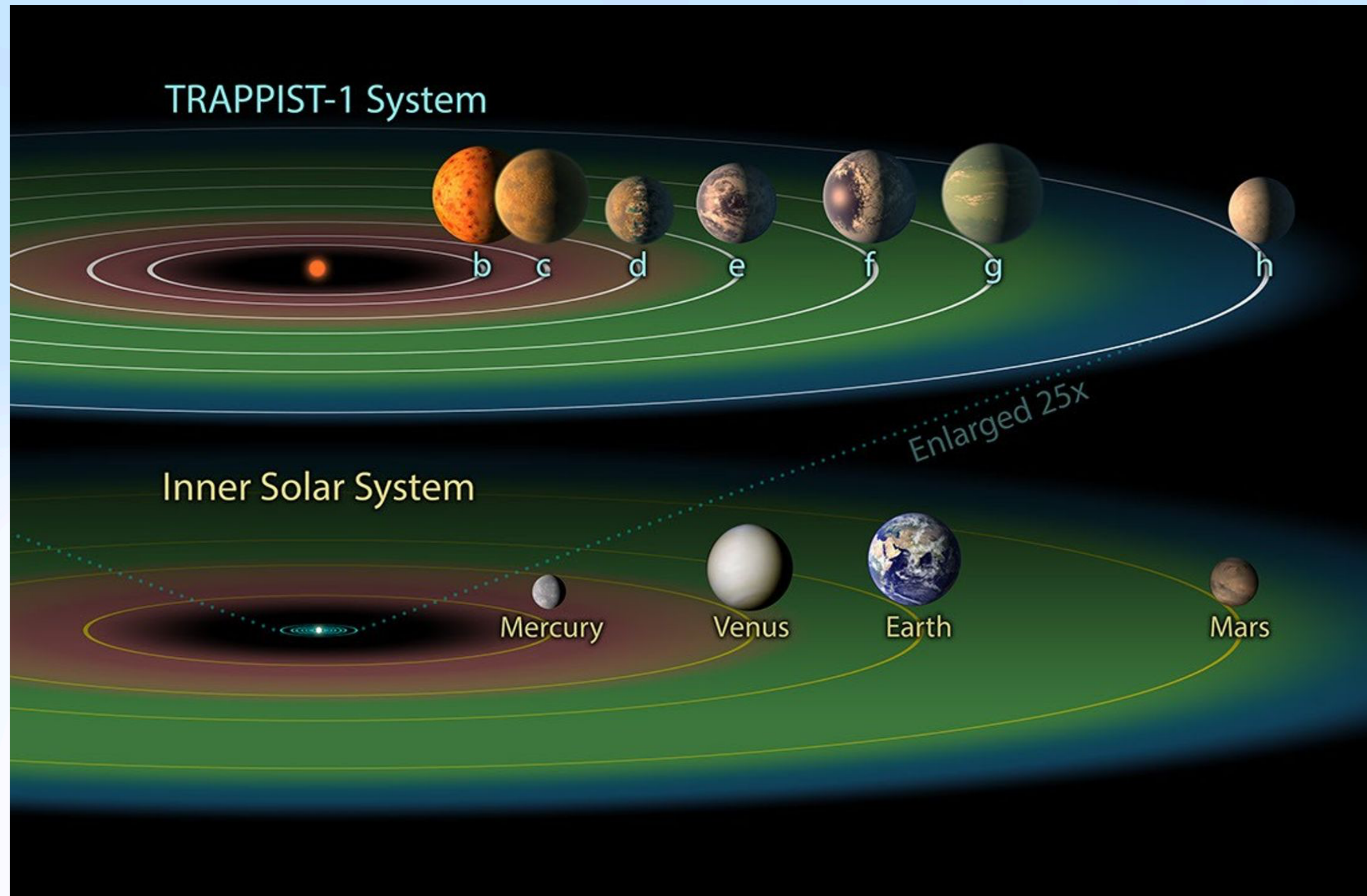
Composition Constraints of the TRAPPIST-1 Planets from Pebble Accretion

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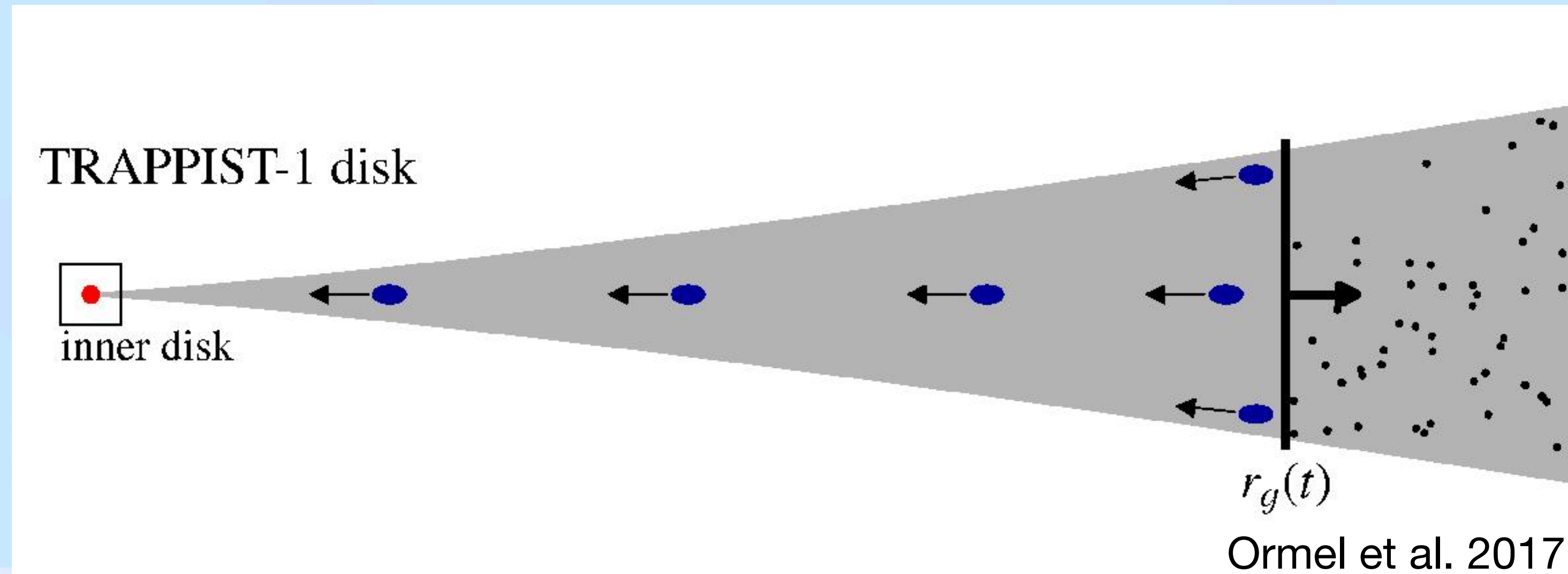
June 12, 2026

TRAPPIST-1

- $M_{\star} = 0.09 M_{\odot}$
- 7 rocky planets between 0.01-0.1 au
- Each planet pair is in a MMR
- Each planet triplet is in a 3BR
- Planets e, f, and g are in the HZ



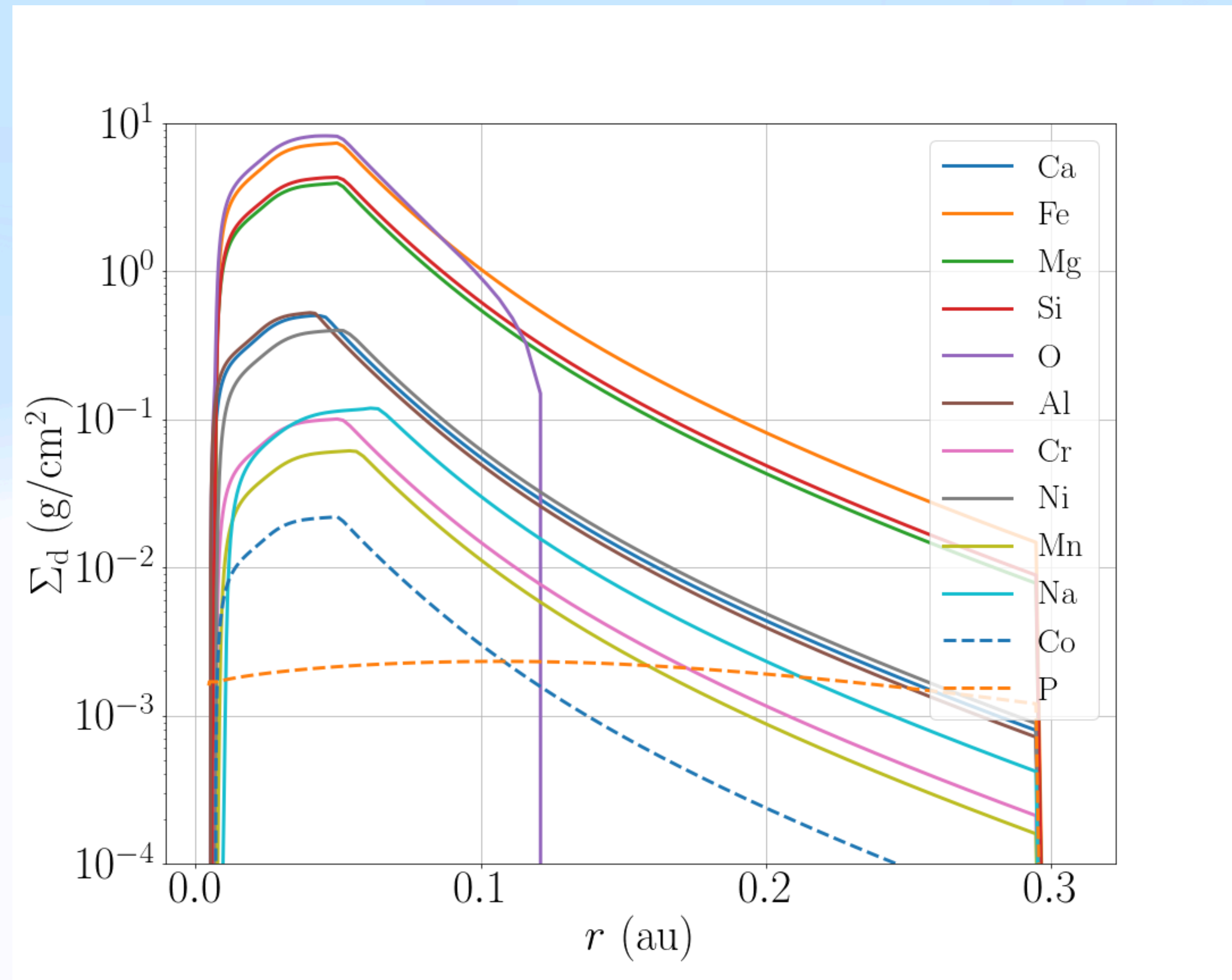
The compactness and resonance structure of the system points towards pebble accretion with inward migration



- Takes place during the gas disk lifetime
- Initial planetesimal population forms exterior to the ice line
- Type I migration
- Mass growth via pebble accretion up to the pebble isolation mass (PIM)
- Eccentricity and Inclination dampening

Can we place constraints on the bulk composition of the TRAPPIST-1 planets by modeling their formation process?

- Build a pebble accretion module for REBOUNDx
- Initialize compositions of planetesimals and pebbles using results from a disk condensation code
- Model the formation of TRAPPIST-1 like systems and track their composition throughout the process
- Feed those results to an interior structure code to compare to observed planet radii



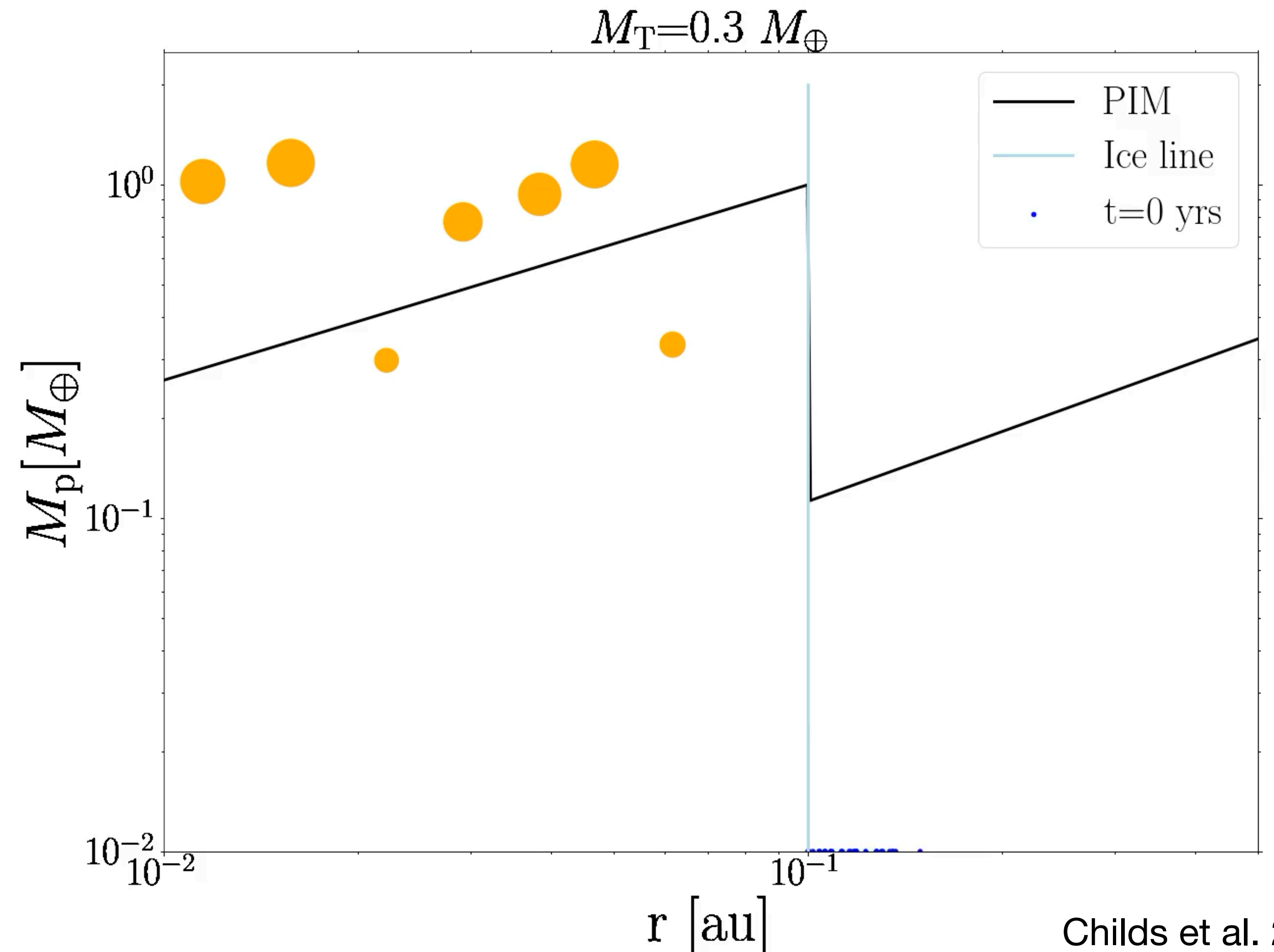
GRAINS code, Cassen 1996, Li et al. 2020, Childs et al. 2023

N-body simulations

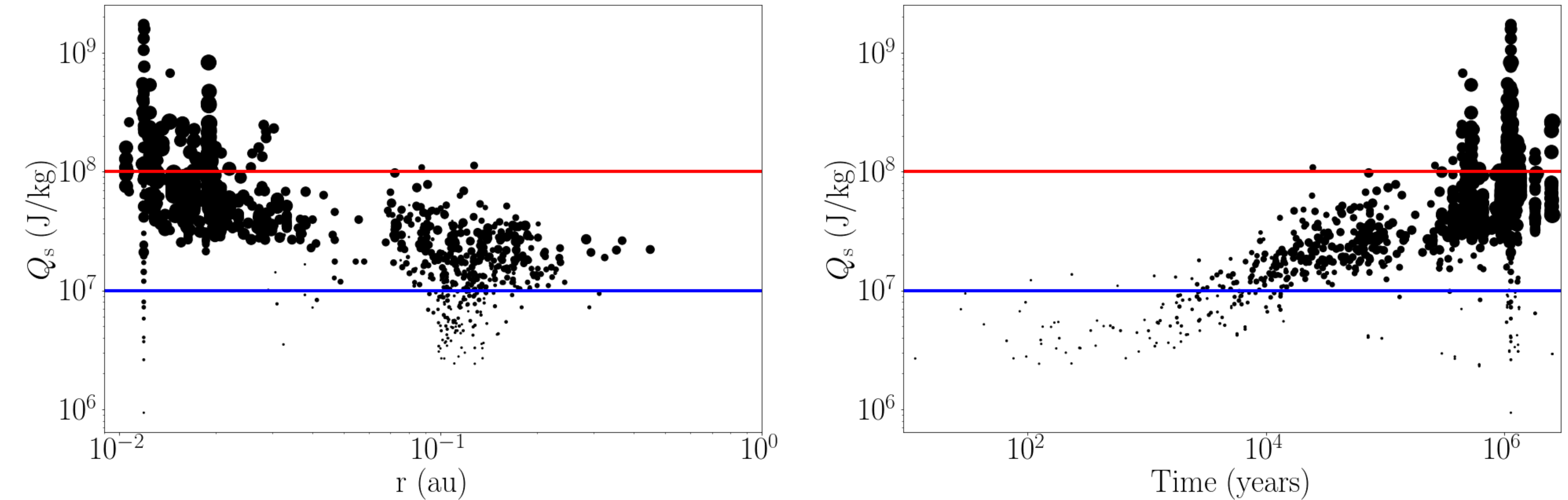
- REBOUNDx and REBOUND with fragmentation
- 30 Mars sized embryos exterior to ice line at 0.01 au
- All bodies have a density of 1.5 g/cm^3
- Planet trap at 0.01 au
- Integrate for 3 Myr
- 100 runs

$$\text{PIM} = (H/r)^3 \sqrt{37.3\alpha + 0.01} \times \left[1 + 0.2 \left(\frac{\sqrt{\alpha}}{H/r} \sqrt{\frac{1}{\tau_s^2} + 4} \right)^{0.7} \right] M_\star$$

Ataiee et al. 2018



Giant impacts still take place!



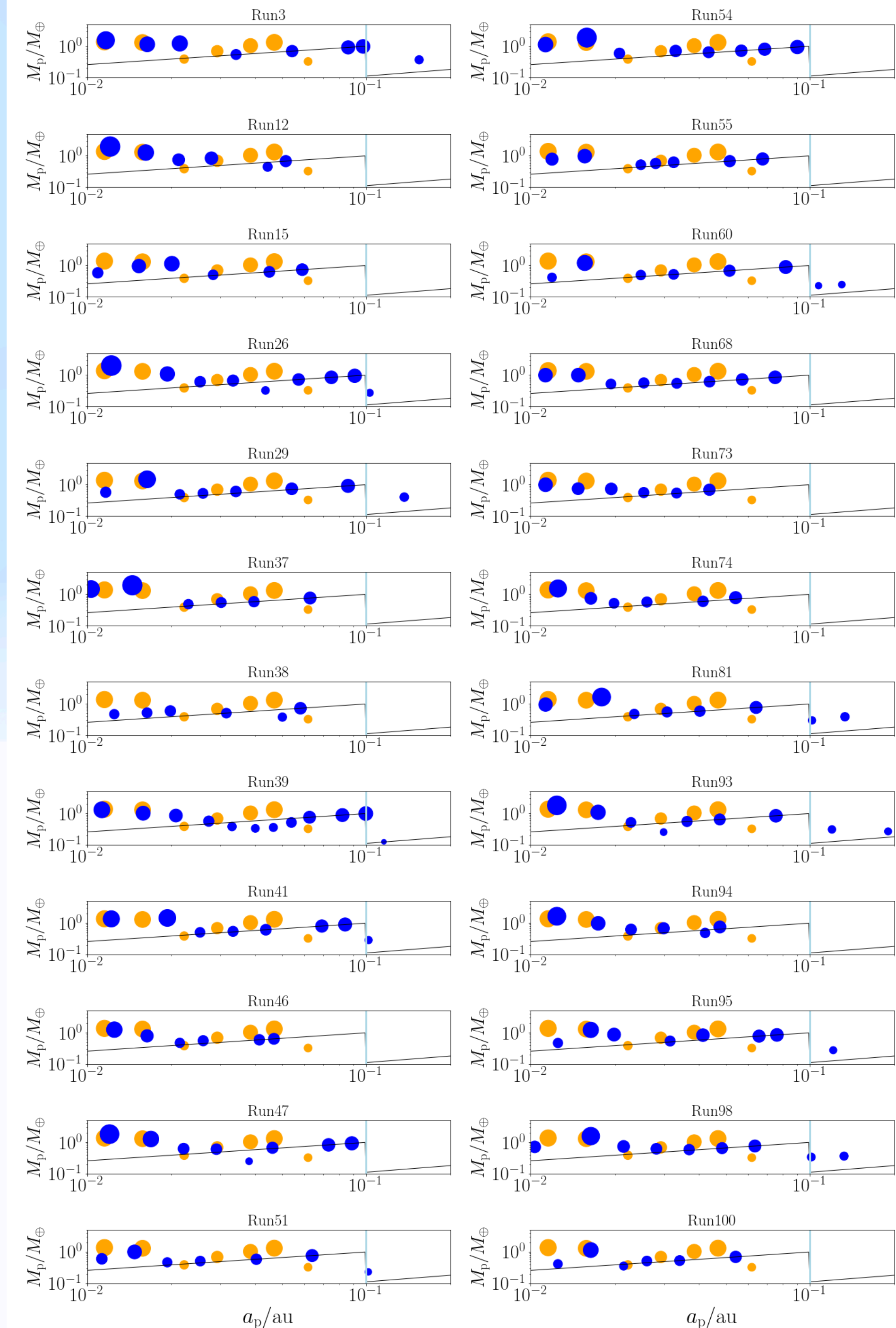
Stewart et al. 2014, Childs et al. 2023

Lower blue line = impact energy to strip an Earth-like atmosphere

Upper red line = impact energy to strip an Earth-like surface ocean

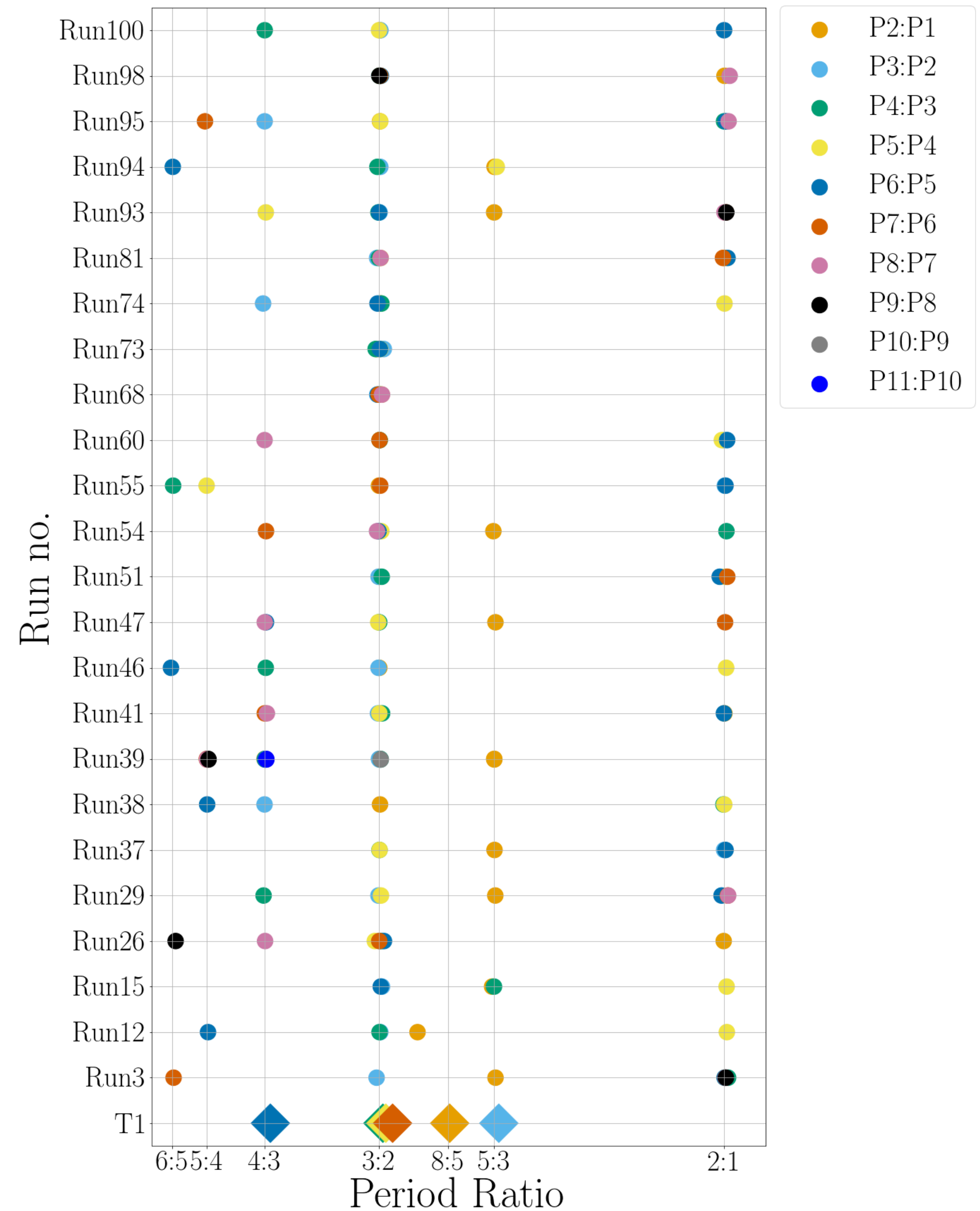
N-body Results

- T1 analogs have at least 6 planets
- 24/100 runs formed T1 analogs
- Fragments seeded at least one planet
- The remaining runs formed fewer, more massive planets



Simulated Resonance Structures

- Nearly all planet pairs found in MMRs
- A simple planet trap can not reproduce the 8:5 MMR



We can also fit for WMF as a free parameter

3-layer model

Planet	Fe/Si	CMF	MMF	WMF
d	0.84	0.283	0.656	$0.061^{+0.023}_{-0.020}$
e	0.84	0.288	0.674	$0.039^{+0.021}_{-0.017}$
f	0.84	0.281	0.663	$0.056^{+0.019}_{-0.015}$
g	0.84	0.274	0.651	$0.075^{+0.020}_{-0.017}$
h	0.84	0.296	0.614	$0.090^{+0.047}_{-0.040}$

2-layer model

Planet	CMF _O	CMF _D	Solution (per cent)
b	$0.20^{+0.06}_{-0.05}$	$0.30^{+0.02}_{-0.02}$	92.0
c	$0.20^{+0.05}_{-0.05}$	$0.30^{+0.02}_{-0.02}$	96.6
d	$0.11^{+0.05}_{-0.05}$	$0.30^{+0.02}_{-0.02}$	96.2
e	$0.18^{+0.05}_{-0.04}$	$0.30^{+0.004}_{-0.01}$	99.5
f	$0.14^{+0.04}_{-0.04}$	$0.30^{+0.02}_{-0.01}$	98.8
g	$0.10^{+0.04}_{-0.04}$	$0.30^{+0.01}_{-0.01}$	96.5
h	$0.05^{+0.04}_{-0.03}$	$0.33^{+0.06}_{-0.04}$	45.0

CMF_O two-layer models in
MAGRATHEA which match 5000 draws
of the observed bulk densities

$$\text{CMF}_D = \text{CMF} / (1 - \text{WMF})$$

Predicted desiccation of T1-b and T1-c has been seen by JWST

- TRAPPIST-1 h likely has an icy shell

Article

Thermal emission from the Earth-sized exoplanet TRAPPIST-1 b using JWST

<https://doi.org/10.1038/s41586-023-05951-7>

Received: 22 January 2023

Accepted: 13 March 2023

Published online: 27 March 2023

 Check for updates

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The TRAPPIST-1 system is remarkable for its seven planets that are similar in size, mass, density and stellar heating to the rocky planets Venus, Earth and Mars in the Solar System¹. All the TRAPPIST-1 planets have been observed with transmission spectroscopy using the Hubble or Spitzer space telescopes, but no atmospheric features have been detected or strongly constrained^{2–5}. TRAPPIST-1 b is the closest planet to the M-dwarf star of the system, and it receives four times as much radiation as Earth receives from the Sun. This relatively large amount of stellar heating suggests that its thermal emission may be measurable. Here we present photometric secondary eclipse observations of the Earth-sized exoplanet TRAPPIST-1 b using the F1500W filter of the mid-infrared instrument on the James Webb Space Telescope (JWST). We detect the secondary eclipses in five separate observations with 8.7σ confidence when all data are combined. These measurements are most consistent with re-radiation of the incident flux of the TRAPPIST-1 star from only the dayside hemisphere of the planet. The most straightforward interpretation is that there is little or no planetary atmosphere redistributing radiation from the host star and also no detectable atmospheric absorption of carbon dioxide (CO₂) or other species.

Summary

- We can place constraints on the bulk composition of the TRAPPIST-1 planets by numerically modeling their formation
- We can couple our constraints with an interior structure model to compare to observed bulk densities to test our predictions
- The compactness, resonance structure and the mass distribution of the TRAPPIST-1 planets points towards a pebble accretion formation channel
- Fragmentation with pebble accretion helps produce high multiplicity systems
- Giant impacts are necessary for reproducing the larger masses of TRAPPIST-1 b and c